



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

**Salem Terminal TO NALLAVALLEIAMMAN
PETROLEUM AG**

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

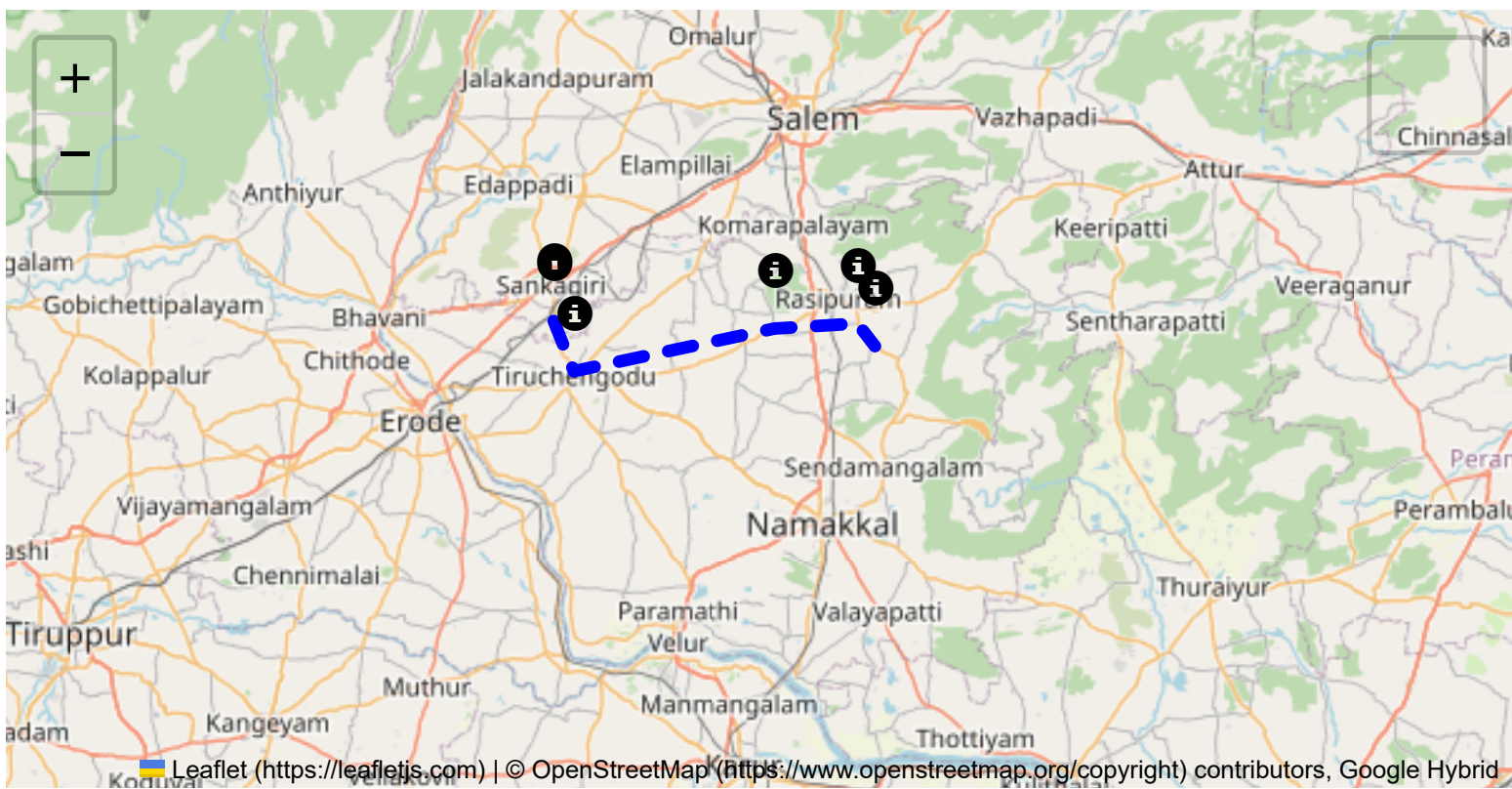
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

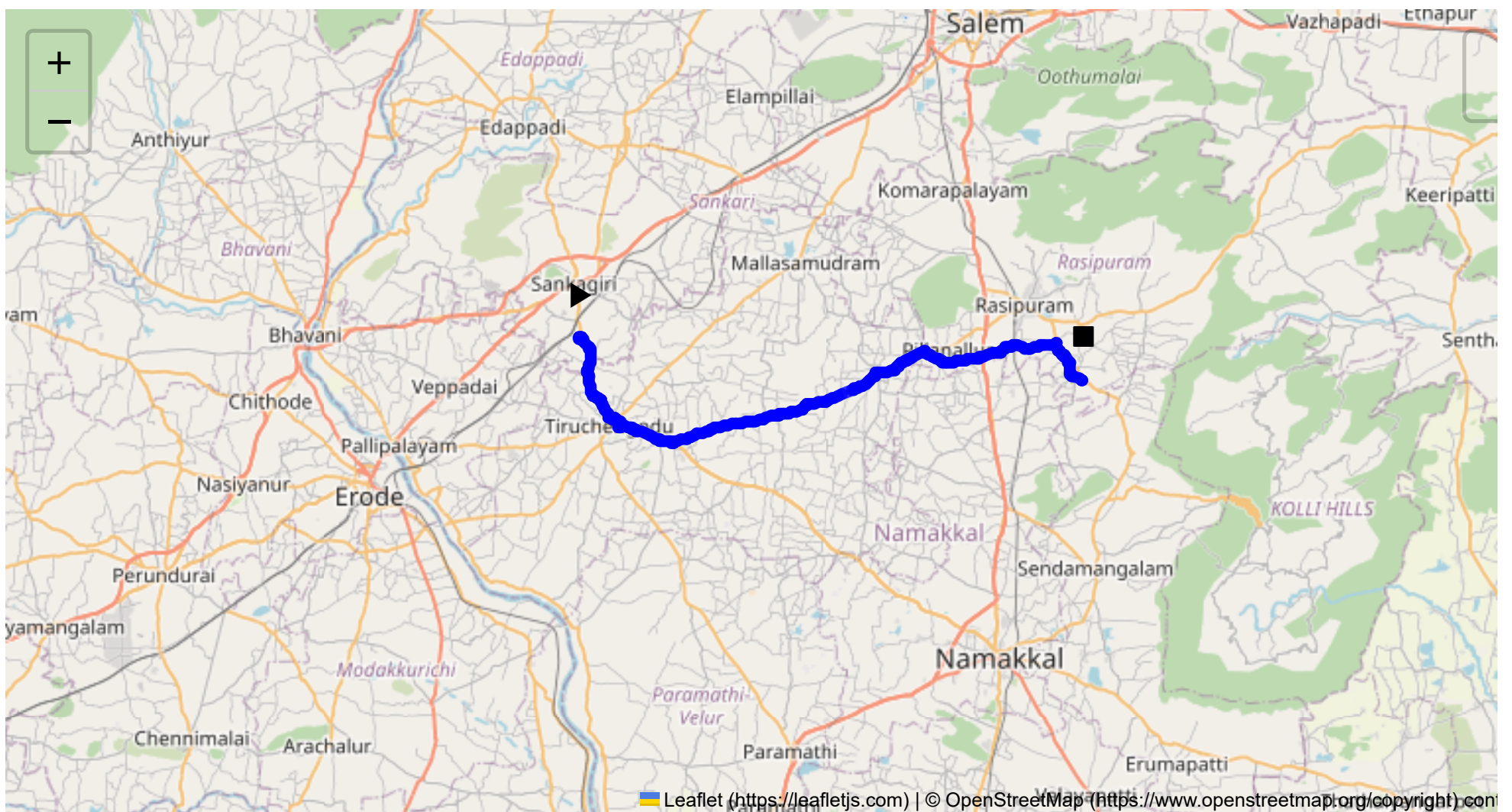
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 48.58 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.409642, 78.225349)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 48.58 kilometers through various towns and rural areas of Tamil Nadu, India. Starting from Sangagiri, it moves through Puthur, Thondikaradu, Ponkurichi, and Thottiyampatty, before reaching Kadiranallur.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Monsoon Season (June-September):** Heavy rainfall can cause flooding and reduce visibility. Landslides might be a concern in elevated areas.
- **Summer (March-May):** High temperatures may affect vehicle performance and road conditions.
- **Winter (November-February):** Generally mild, but fog can reduce visibility during early mornings and late nights.

3. Analysis of Traffic Patterns

- **Peak Hours:** Expect heavy traffic during school hours and typical office hours (8 AM - 10 AM and 5 PM - 7 PM) near urban areas.
- **Congestion-Prone Areas:**
 - Near Girls High School and Bharat Petrol Bunk in Thondikaradu.
 - Market areas in Rasipuram and Tiruchengode.

4. Assessment of Road Quality and Infrastructure

- **Road Conditions:** Varying from well-paved to uneven in rural sections.
- **Infrastructure:** Limited in some rural areas, with narrow roads and limited safety barriers. The stretch through Tiruchengode and Rasipuram is better maintained.

5. Suggestions for Alternative Routes for Emergencies

- An alternative route via NH544 could be considered in case of roadblocks or construction. This would bypass some rural impediments but might increase travel distance.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Time Restrictions:** Transport of hazardous materials may be restricted during daytime in populated areas.
- **Permit Requirements:** Special permits are likely required for carrying hazardous substances.

7. Overview of Historical Incidents

- Minimal incidents reported involving hazardous materials, but occasional accidents have occurred due to driver error or poor road conditions.

8. Environmental Considerations and Sensitive Areas

- **Proximity to Schools and Residential Areas:** Requires careful navigation and reduced speeds.
- **Agricultural Zones:** Be mindful of farm vehicles and animals on rural paths.

9. Analysis of Communication Coverage

- Generally good network coverage, but potential dead zones exist in some rural segments, notably between Puthur and Thondikaradu.

10. Estimated Emergency Response Times

- **Urban Areas (Tiruchengode, Rasipuram):** 15-30 minutes.
- **Rural Areas:** May extend to 45 minutes or longer due to accessibility issues.

12. Overall Summary of Risk Assessment

The route includes both urban and rural segments with varying road conditions, requiring alertness and cautious driving, particularly for trucks carrying hazardous materials. Key risks include weather-related hazards, traffic congestion, and limited infrastructure in rural areas. Planning for alternative routes and understanding local regulations are crucial for safe transportation. Emergency response times vary significantly, which should be considered in route planning.

Overall, while the route is generally passable, caution and prior preparation are strongly advised, especially during adverse weather conditions or peak traffic times.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	High	11.38190, 77.90254	15 KM/Hr
7	Turn	Medium	11.38182, 77.90259	30 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Turn	Medium	11.41447, 78.08063	30 KM/Hr
10	Turn	Medium	11.41468, 78.08072	30 KM/Hr
11	Turn	Medium	11.41475, 78.08087	30 KM/Hr
12	Turn	High	11.42964, 78.11587	15 KM/Hr
13	Turn	Medium	11.42956, 78.11590	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	Medium	11.42450, 78.15278	30 KM/Hr
15	Blind Spot	Blind Spot	11.43540, 78.20751	10 KM/Hr
16	Turn	Medium	11.43040, 78.20860	30 KM/Hr
17	Turn	Medium	11.42751, 78.21298	30 KM/Hr
18	Turn	High	11.42171, 78.21703	15 KM/Hr
19	Blind Spot	Blind Spot	11.41011, 78.22543	10 KM/Hr
20	Blind Spot	Blind Spot	11.40969, 78.22524	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38182, 77.90259



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41447, 78.08063



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41468, 78.08072



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41475, 78.08087



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42956, 78.11590



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42450, 78.15278



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.43540, 78.20751



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43040, 78.20860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42751, 78.21298



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42171, 78.21703



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41011, 78.22543



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.40969, 78.22524
